

Question				Marking details			Marks available					
							AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Hazard	Risk	Control measure		2		2		
				Apparatus is hot	Burning {skin/ yourself} when {moving / touching} {apparatus/ boiling tube} (1)	Allow apparatus to cool / use tongs (to hold boiling tube)/ wear <u>heatproof</u> gloves (1)						
		(ii)	I	Use a balance / (weighing) scales				1		1		1
			II	Any two (×1) from: Volume of water (1) Distance of {food/ flame} from test tube (1) {Mass/ weight} of food (1) Starting temperature (1)					2	2		2
			III	{Measure/ take/ record} {temperature/ °C} (of water using a thermometer) before and after burning (1) Calculate the difference in temperature (1) End temperature – start temperature = 2 marks					2	2		2
		(iii)		{Total/ maximum/ all} energy (content) released (or equivalent wording)				1		1		1
	(b)	(i)		28.0 (J/g) = 2 marks 28 = 1 mark $\frac{252}{9} = 1$ mark				1 1		2	2	
		(ii)		Plain (allow ecf if calculation error made) as highest energy content <u>per gram</u>					1	1		2
				Total for Question 4			0	6	5	11	2	10